

Increase Your Tendon Strength

Six Methods to Improve Tendon Strength...Muscle Growth

1. Isometrics/Isometronics
2. Slow eccentrics
3. Isolation Movements within problem areas - Bodybuilding...
4. Higher force plyometric movements...with precise control
5. Mobility Methods
 - a. Tack and floss
 - b. Manipulation and scraping
 - c. Stretching...shorter duration to long duration
6. Nutrition

API Diagnosis

I. Assess

- A. What causes pain? Movement/positions/intensity
- B. What tendencies lead to this issue? Foot positions - duck footed?/shoulder position etc.
 - 1. Plopping on jumps/heel collapse/too wobbly shoulders...lack of cocontractions

II. Plan

- A. Warm ups?
- B. Training?
- C. Cooldowns?
- D. Changes to the program...

III. Implementation

- A. Make it part of the system
- B. Mobility is KEY to athletic success

Assessment

I. Consistent Joint/Tendon Issues

A. Throwers

1. Wrist
2. Elbows and Shoulders
3. Knees
4. Groin
5. Ankle

B. Sprinters

1. Ankle - achilles
2. Knees
3. Hips



Assessment

- A. Jumpers
 - 1. Ankle - achilles
 - 2. Knee - Jumpers knee (patellar tendinopathy)
 - 3. Hip
- B. Distance Runners
 - 1. Ankle - Achilles
 - 2. IT Band Stiffness
 - 3. Runners Knee
 - 4. Hip Flexor
 - 5. Plantar Fasciitis



Assessment

I. Types of Tendon Pain

- A. Chronic from stiffness, adhesions
 - 1. Improve warm ups
 - 2. Analyze any weird movements in technical sport
 - 3. Focus on resistance training/proper load
 - 4. How consistent are they with mobility and recovery
- B. Stinging pain from microtears or acute issues
 - 1. Generally, recommended to get imaging if possible



Planning

I. Planning the Daily Application

- A. Warm ups
- B. In Drills as Potentiation
- C. Resistance Training
- D. Recovery

II. Weekly Split

- A. Daily Cycle of Warm Ups and Focus Points
- B. Target Areas depending on the lifting focus
- C. Mobility for recovery based on following day training



Planning

- I. Dial in Two Problem Areas within the Event
- II. Two Problem Areas Get Concurrently Treated...I.E.
 - A. Day 1
 - 1. Warm ups - Knee focused
 - 2. Lift - Eccentric focused on knee
 - 3. Recovery - Mainly shoulder focused
 - B. Day 2
 - 1. Warm ups - Shoulder Focused
 - 2. Lift - Eccentric focused on Shoulder
 - 3. Recovery - knee or back focused



Planning

I. Yearly or Seasonal Preparation

A. Higher Volume - Early Season

1. Use more low intensity jumps
2. Pauses in jumps and resistance work
3. Use more pauses and isometrics including during compound movements
4. Hypertrophy/Isolation work on possible issue points

B. Peaking

1. Remove the volume and pauses
2. Remove hypertrophy/isolation work

Implementation into a Yearly System of Training

- I. Warm Ups
 - A. Isometrics
 - B. Short duration mobility work/skilled tack and floss or scraping
- II. Technical Work and Drills
 - A. Isometrics/Pauses
- III. Resistance Based Training
 - A. Isometrics
 - B. Slow Eccentrics
 - C. Isolation Movements
- IV. Plyometric Work
 - A. Isometrics
 - B. Low intensity/barefoot
- V. Mobility
 - A. Longer duration
 - B. Static Positions



Warm Ups...

- I. Strategy Deployed
 - A. Isometrics
 - B. Short duration mobility work/skilled tack and floss or scraping
- II. Ankle/Foot
 - A. Toe walks
 - B. PVC Walks
 - C. Heel to Toe Rocks
- III. Knee
 - A. PVC Walks
 - B. Wall sit/couch stretch/split squat long and slow
- IV. Hip/IT
 - A. Hip 90/90 - couch stretch - foam roll - chair stretch
- V. Elbow/Wrist
 - A. Wrist flexion hold
 - B. Lengthened band tension - extended band tension



Technical Work and Drills

I. Strategy Deployed

- A. Isometrics
- B. Pauses

II. Ankle/Foot

- A. 2-3 active drills then pause drills/Wall Pushes for time

III. Knee

- A. Pause and hold in positions that may be painful - SLOW drills and flow = batting practice

IV. Hip/IT

- A. Much harder here unless work is lateral or leading to hip flexion position

V. Elbow/Wrist

- A. n/a

Resistance Work

- I. Strategy Deployed - Only do if feeling improves during session
 - A. Isometrics
 - B. Slow Eccentrics
 - C. Isolation exercises
- II. Ankle/Foot
 - A. Heel unsupported split squats
 - B. PVC Walks and split squat PVC's
 - C. Tibialis Curls/Calf raises with Iso holds and slow eccentrics
- III. Knee
 - A. Slow 4-8 second eccentric squats and split squats slow eccentric with iso holds
 - B. Spanish Squats
 - C. Pause squats/slow eccentric pause front squats
 - D. Leg Extensions
 - E. Sled work/backward hill walks/backward stair climbs

Resistance Work Continued

I. Hip/IT

- A. Cossack Squats slow eccentrics with isometric hold
- B. Overhead lunges slow eccentric with iso holds
- C. Curtsy Squats or Curtsy step ups with iso holds
- D. Banded Back Extensions/Glute Ham raises
- E. Hanging Leg Raises pause iso hold
- F. Side Band Walks/Clam Shells pausing

II. Elbow/Wrist

- A. Forearm Roller - Reverse curls slow eccentric - rotational banded curls
- B. Bicep curls slow eccentric - incline curls very slow eccentric
- C. SLOW eccentric elbow flexion into ISOMETRIC hold at the lockout

5 S's Protocol

- I. Slow Jumps
- II. Stop
- III. Stutter Jumps
- IV. Speed
- V. Sport Specific



Plyometric Training

I. Strategy Deployed

- A. Isometrics
- B. Low intensity/barefoot

II. Ankle/Foot

- A. Rapid PVC walks
- B. Barefoot low intensity jumps - Pogo hops/jump ropes/easy skips - Double leg jump to single leg land on low boxes
- C. Double bounce hops to jump

III. Knee

- A. Depth drop to PAUSE landing
- B. Jump lunges pause and hold split/box jump pause landing one leg hold
- C. Hurdle hops pause each landing 2-5 seconds
- D. Stair jumps with diminished load

Plyometric Training

I. Hip/IT

- A. Jump lunges
- B. Rotational Jump lunges
- C. Side to Side jumps with pauses
- D. Bound to Turn and Pause
- E. Pause Heiden Jumps

II. Elbow/Wrist

- A. Double bounce push ups
- B. Clap Push Ups
- C. Push ups to boxes
- D. Explosive Pull ups squeeze at the top



Mobility

I. Ankle/Foot

- A. PVC Split squat hold
- B. Ankle cranking
- C. Knee To Wall stretch

II. Knee

- A. Pigeon Pose
- B. Side quad stretch
- C. Hamstring stretches seated and lying

III. Hip/IT

- A. Half kneeling hip flexor
- B. Lying figure 4
- C. Brettzel
- D. Hip 90/90
- E. Pause Squat holds

IV. Elbow/Wrist

- A. Isometric wrist hold and elbow extension stretches
- B. Rotational movements with racket or object



Nutrition/Recovery

I. Foundations are Key

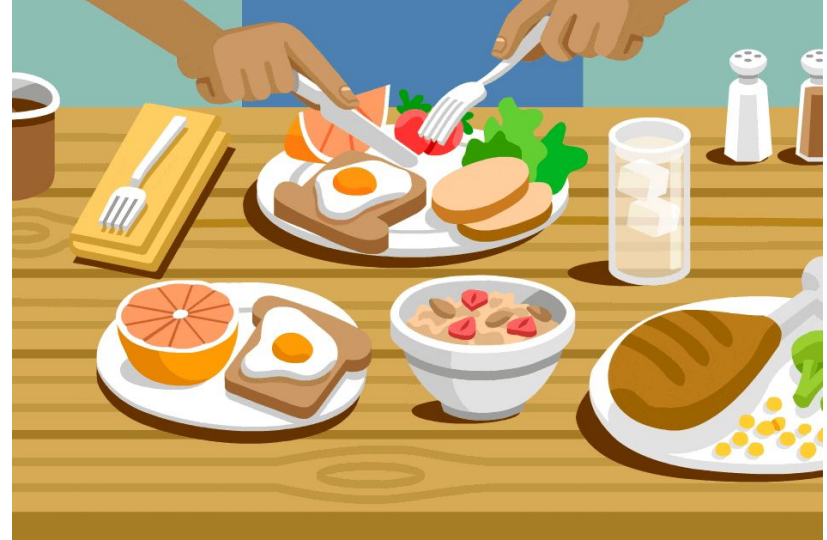
- A. Moderate to high protein
- B. Carbs if you feel good on carbs
- C. Fats if you feel good on fats

II. Avoid

- A. Trans fats
- B. Highly heated oils
- C. Any potential allergens for you specifically

III. Supplements that MIGHT Help

- A. Creatine
- B. High quality fish oils
- C. Collagen MIGHT help if supplemented with Vitamin C



Recap

- I. Know the 6 methods - Isometrics/Eccentrics/isolation/plyometrics/mobility and nutrition
- II. Know the event and the regions PRONE to injury with the individual
- III. Know the strategy to use at WHAT point in a session
- IV. Implement over a long period of time for success and prevention

